REAL-TIME PROPULSION HEALTH MONITORING AND FAULT-TOLERANT CONTROL

for an Autonomous Marine UAV

Project Report

An STM32 NUCLEO-C031C6 based embedded prototype for monitoring propulsion health using RPM, motor current, temperature and vibration, with real-time fault classification and protective control.

Platform: Wokwi simulation
Controller: STM32 NUCLEO-C031C6
Display: SSD1306 I²C OLED
Motion sensor: MPU6050

1. Abstract

This project presents a real-time propulsion health monitoring and fault-tolerant control prototype for an autonomous marine UAV. The system uses an STM32 NUCLEO-C031C6 microcontroller to acquire simulated RPM, motor-current and temperature measurements, together with dynamic vibration information from an MPU6050. A lightweight sensor-fusion algorithm classifies propulsion conditions as normal, warning, critical or emergency. The resulting condition is presented through an OLED display and status LEDs, while a buzzer and servo provide simulated protective control actions. The prototype was implemented and validated in Wokwi, demonstrating sensor acquisition, vibration fault detection, event counting, status recovery and integrated control behavior.

2. Problem Statement

Propulsion faults such as propeller obstruction, excessive vibration and motor overheating can reduce the reliability of an autonomous marine UAV. A monitoring system must identify abnormal conditions quickly and initiate an appropriate protective response. The project therefore focuses on combining multiple propulsion indicators rather than depending on a single sensor.

3. Objectives

• Monitor propulsion RPM in real time.
• Estimate motor current and temperature using simulated analog inputs.
• Measure dynamic acceleration changes as a vibration indicator.
• Fuse sensor information for propulsion-fault classification.
• Provide warning, critical and emergency responses.
• Display live system condition through an OLED.
• Maintain fault-event counters for basic maintenance awareness.

4. System Architecture

Input / Sensor	STM32 Processing	Output / Action
KY-040 — RPM	Sensor acquisition	SSD1306 OLED
Potentiometer — Current	Sensor fusion	Green / Yellow / Red LEDs
Potentiometer — Temperature	Fault classification	Buzzer
MPU6050 — Vibration	Health assessment	Servo protection control

5. Hardware and Pin Configuration

Component	Pin / Signal	STM32 Connection
KY-040	CLK	D2 / PA10
KY-040	DT	D3 / PB3
KY-040	SW	D4 / PB10
Current input	SIG	A0 / PA0
Temperature input	SIG	A1 / PA1
MPU6050	SDA	D14 / PB9
MPU6050	SCL	D15 / PB8
OLED	SDA	D14 / PB9
OLED	SCL	D15 / PB8

Component	Pin / Signal	STM32 Connection
Green LED	Control	D5
Yellow LED	Control	D6
Red LED	Control	D7
Buzzer	Control	D8
Servo	Signal	D9

6. Fault Detection Logic

The controller applies priority-based fault classification. Emergency conditions are handled before critical and warning conditions so that severe propulsion conditions receive the strongest protective response.

Condition	Threshold	Classification	Action
Normal	Within limits	NORMAL	Maintain propulsion
Propeller obstruction	RPM < 30 and current ≥ 6 A	WARNING	Reduce propulsion RPM
High vibration	Vibration ≥ 300 mg	WARNING	Check propeller / bearing
High temperature	70–79 °C	WARNING	Reduce propulsion load
Critical temperature	80–89 °C	CRITICAL	Thermal protection
Severe obstruction	RPM < 15 and current ≥ 8 A	EMERGENCY	Emergency throttle reduction
Overheating	Temperature ≥ 90 °C	EMERGENCY	Emergency shutdown

7. Vibration Measurement Approach

The MPU6050 provides raw acceleration values. The prototype compares consecutive samples on the X, Y and Z axes, takes the absolute differences and combines them into a dynamic-change indicator. The result is converted to an approximate mg-scale value using the 16384-count-per-g reference used for the selected accelerometer range. A value of 300 mg or above is treated as a high-vibration warning. This is a lightweight dynamic vibration indicator for the simulation and should not be interpreted as a laboratory-grade RMS vibration measurement.

8. Fault Event Monitoring

The final prototype maintains counters for obstruction, overheating, vibration and emergency events. Counters increment on the transition into a fault condition rather than continuously on every program loop. This provides a simple event-history mechanism that can support future maintenance and predictive-monitoring extensions.

9. OLED and Actuator Response

The SSD1306 OLED shares the I²C bus with the MPU6050. The output layer provides live monitoring and fault indication. Green indicates normal operation, yellow indicates a warning, and red with the buzzer indicates critical or emergency conditions. The servo is used as a simulated propulsion/throttle actuator: approximately 90° for normal operation, 60° for warning, 30° for critical protection and 0° for emergency response.

10. Validation Results

Test	Observed result	Status
System startup	MPU6050 READY; OLED READY; SYSTEM READY	PASS
Normal operation	Status remained NORMAL	PASS
Vibration fault	2149 mg → WARNING	PASS
Vibration recovery	0 mg → NORMAL	PASS
Vibration event counter	Counter changed from 0 to 1	PASS
RPM input	Observed 6, 9 and 12 RPM values	PASS

11. Advantages

• Multi-parameter propulsion monitoring instead of single-sensor detection.
• Priority-based fault classification.
• Automatic protective response through simulated control actuation.
• Local OLED and LED/buzzer indication.
• Fault-event counting for basic maintenance awareness.
• Suitable for rapid embedded-system prototyping and demonstration.

12. Limitations

The prototype is validated in Wokwi simulation. The current and temperature channels are represented by potentiometers, the KY-040 provides a simulated/manual RPM input, and the servo represents a simulated propulsion-control actuator. The vibration calculation is a dynamic-change indicator rather than a certified vibration-analysis measurement. Therefore, the prototype demonstrates the embedded monitoring and control architecture but is not a flight-certified or marine-certified propulsion controller.

13. Future Scope

Future development can replace the simulated inputs with calibrated current, temperature and RPM sensors; implement RMS, peak-to-peak and trend-based vibration analysis; add non-volatile fault logging; transmit telemetry to a ground station; and use collected propulsion data for machine-learning-based fault prediction. Hardware-in-the-loop and physical propulsion testing would be required before any real autonomous-vehicle deployment.

14. Conclusion

The project successfully demonstrates a complete embedded propulsion-health monitoring and fault-tolerant control workflow using the STM32 NUCLEO-C031C6. RPM, current, temperature and vibration information are combined to identify propulsion abnormalities. The validated prototype detects a high-vibration event, records the event, and returns to normal status when the vibration condition clears. OLED monitoring and actuator responses provide a practical operator-facing interface. The resulting architecture forms a strong foundation for future real-sensor, telemetry and predictive-maintenance development.

Project Status: Core prototype validated in Wokwi.

The report intentionally distinguishes simulated inputs and prototype behavior from real-world marine UAV deployment.